King Faisal University

College of Computer Sciences and Information Technology

Embedded systems

Project title:

Machine Guardian

Members:

Fehaid Ali Alshishaee – 222403548

Hassan Ibrahim Rawagah – 222416197

Ali Abdullah Albaik – 223036886

Ali Jawad Al-Mutawah - 222424050

Dr: Rama Sami

Abstract

This report details the development of "Machine Guardian," an integrated IoT system designed to enhance industrial safety, access control, and energy management. Utilizing the ESP32 microcontroller as a central logic unit, the system combines Near Field Communication (NFC) for identity verification with non-invasive current sensing (SCT-013) to monitor heavy machinery. The architecture ensures that equipment operation is restricted to authorized personnel while providing real- time data on power consumption and operational efficiency. Key features include a locally-hosted web dashboard for data visualization, automated Telegram alerts for unauthorized usage, and an event-driven state machine to ensure deterministic behavior. Experimental results demonstrate the effectiveness of the "Silencing Protocol" in reducing electromagnetic interference during analog sampling and the reliability of True RMS calculations for accurate power logging.

Introduction

The Machine Guardian is an integrated IoT solution designed for industrial equipment monitoring, access control, and energy management. Utilizing the ESP32 microcontroller, the system serves as a "smart gatekeeper" for heavy machinery. By combining NFC (Near Field Communication) for identity verification and non-invasive Current Sensing (SCT-013), the system ensures that only authorized personnel can operate equipment while providing real-time data on power consumption and operational efficiency.
The system is further enhanced by a locally-hosted web dashboard and automated Telegram alerts, providing supervisors with immediate visibility into unauthorized usage or equipment status.

Problem Statement

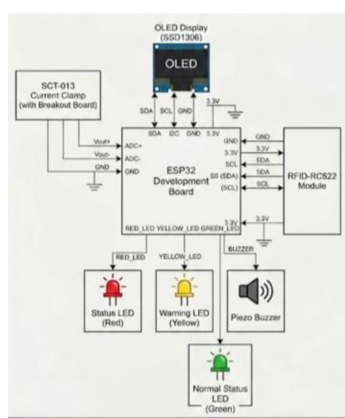
Heavy machinery in industrial settings is often prone to unauthorized operation, which poses significant safety risks and leads to unmonitored energy expenditure. Traditional monitoring methods frequently suffer from: (1) Lack of Access Control, making it difficult to ensure only certified personnel operate specific machines; (2) Inaccurate Sensing due to ambient noise and electromagnetic interference (EMI); and (3) Delayed Notifications in the event of unauthorized use.

System Architecture

The firmware acts as the central logic unit for a multi-modular hardware stack. Its primary components include:
Access Control: A PN532 NFC module validates user credentials against a local database stored in the ESP32's Non-Volatile Storage (NVS).
Sensing & Math: The SCT-013 sensor captures AC current through a burden resistor. The software performs a true RMS (Root Mean Square) calculation to determine actual amperage.
User Interface: A 0.91" SSD1306 OLED displays status, while a buzzer and a three-stage LED system (Red/Yellow/Green) provide visual and audible feedback.
Connectivity: The device operates as a web server, hosting a sophisticated JavaScript-based dashboard for data visualization and employee management. It also functions as a Telegram client for remote notifications.

Hardware Design

The hardware selection focuses on non-invasive installation and low-latency processing. Key components include the ESP32 for dual-core processing, the SCT-013 current sensor for non-contact detection, and a feedback system comprising a 0.91" SSD1306 OLED and a 3-stage LED (Red/Yellow/Green) to indicate machine states.



Software Design

The firmware utilizes an Event-Driven State Machine to maintain high reliability. The logic transitions through states: IDLE, AUTHENTICATED, RUNNING, and UNAUTHORIZED based on NFC verification and current thresholds.

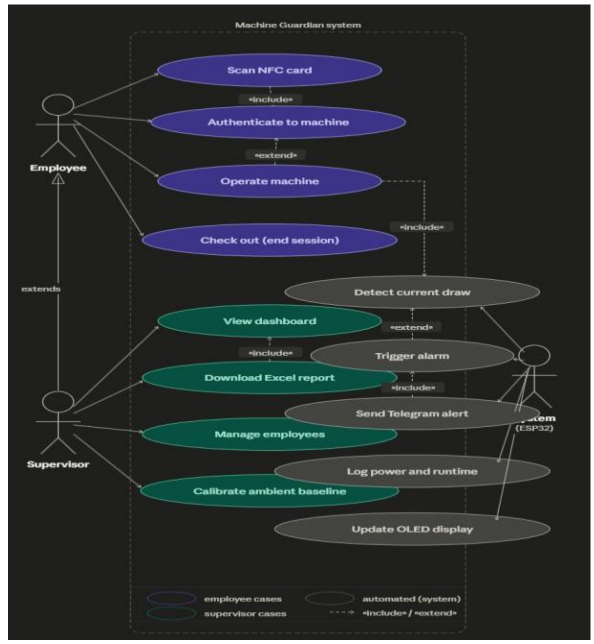
A. Step-by-Step Algorithm

1. Initialize I2C, SPI, and Wi-Fi; load employee records.
2. Wait for NFC scan while monitoring baseline current.
3. Validate UID against local NVS database.
4. Open a 30-second window for machine start if authorized.
5. Halt I2C bus (Silencing Protocol) and take 2000 samples.
6. Calculate True RMS current.
7. Trigger alarm and Telegram alert if current > 1.0A without auth.

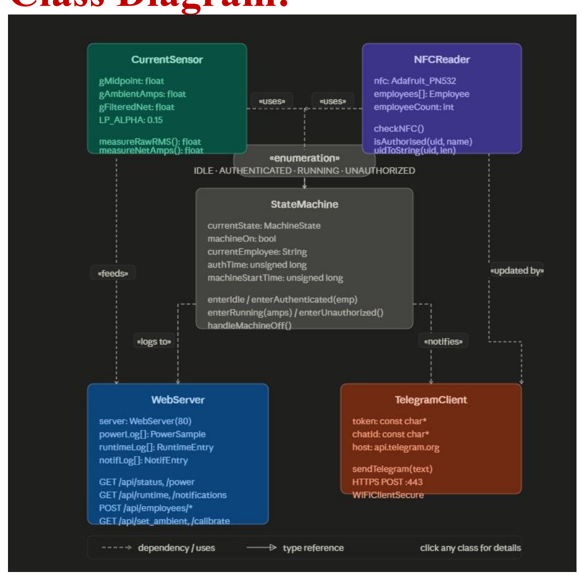
B. Flowchart Description

The system flow initiates at the NFC scan. A decision branch validates the user; success triggers the operational window. Concurrently, the current sensing loop monitors for draw. If draw exceeds the threshold without a valid session, the UNAUTHORIZED state is entered, triggering the alarm system.

Use case Diagram:



Class Diagram:



Methodology

The firmware is structured using an Event-Driven State Machine architecture to ensure high reliability and low-latency response times.
A. Signal Processing & Interference Mitigation
Because the ESP32's Analog-to-Digital Converter (ADC) is sensitive to high-frequency noise, the methodology includes a "Silencing Protocol." During current sampling, the firmware temporarily halts the I2C bus (Wire.end()) to prevent the OLED and NFC modules from injecting electromagnetic interference into the analog readings.
The current is calculated using the following logical flow:
Sampling: 2000 samples are taken over a specific window.
Offset Correction: A dynamic midpoint (bias) is tracked using an Exponential Moving Average (EMA) to account for thermal drift in the DC bias circuit.
RMS Conversion: The raw samples are squared, averaged, and then the square root is taken to obtain the effective current.
B. State Management
The system transitions through four primary states:
IDLE: Machine is off; waiting for a valid NFC tag.
AUTHENTICATED: A valid tag is scanned; a 30-second window is granted to start the machine.
RUNNING: The sensor detects current above the threshold (1.0A); runtime and power logs are updated.
UNAUTHORIZED: Current is detected without a prior NFC scan; the system triggers an alarm and sends a high-priority Telegram alert.
C. Data Persistence & Analytics
Employee records and calibration data are saved using the Preferences library, ensuring the device remains functional after a power cycle without needing an external database. The web dashboard uses Chart.js for client-side rendering of power logs and SheetJS to allow users to export the RuntimeEntry logs into Excel format.

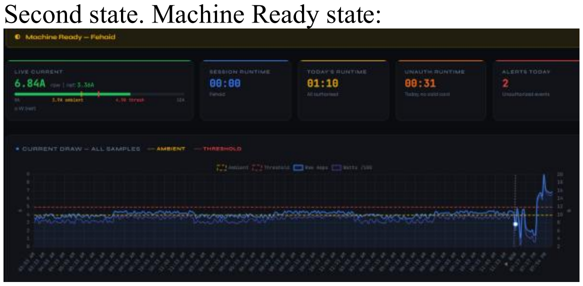
Results and Discussion

The "Silencing Protocol" proved critical, as I2C bus activity previously injected noise into analog readings. By pausing the bus during sampling, a stable current baseline was achieved.

Testing

First state. Idle state:

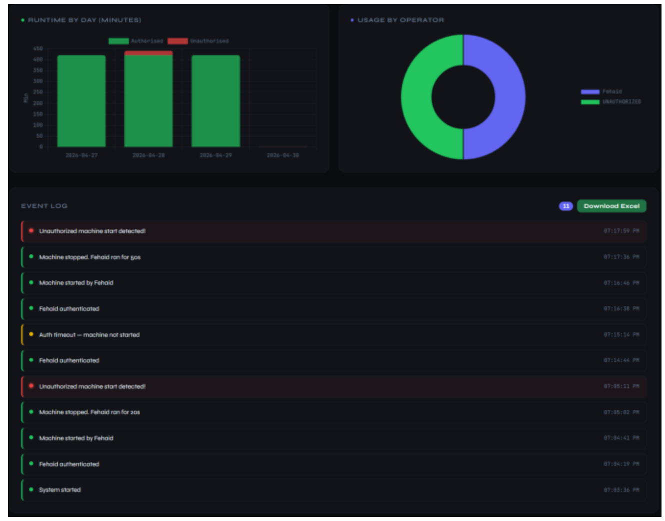
Second state. Machine Ready state:



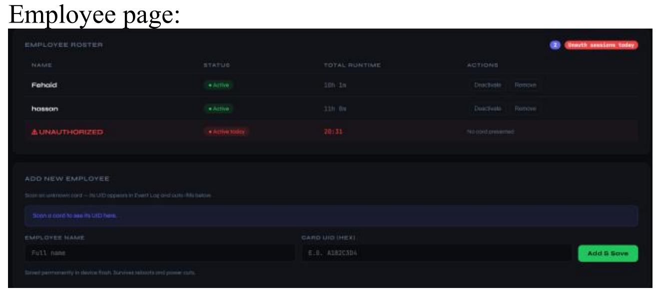
Third state. Running state:

Fourth state. Unauthorized Access state:

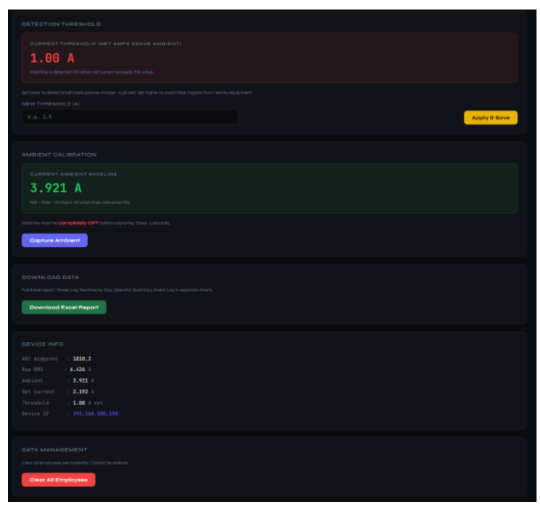
Usage charts & Event log



Employee page:



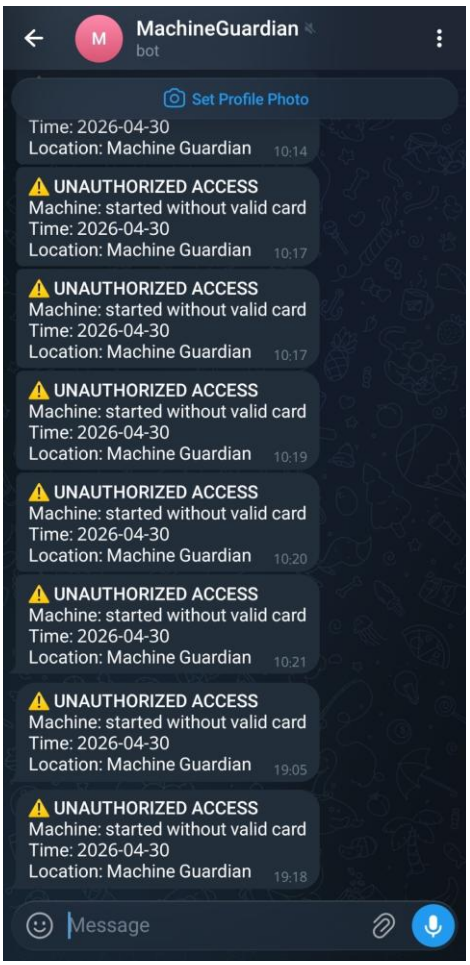
Setting page:



Offline downloaded Excel sheet:

Timestamp	Date	Time	Raw Amps	Net Amps	Watts
30/04/2026, 19:04:46	30/04/2026	07:04 PM	2.82	0	620
30/04/2026, 19:04:51	30/04/2026	07:04 PM	4.76	0.839	1046
30/04/2026, 19:04:57	30/04/2026	07:04 PM	3.48	0	765
30/04/2026, 19:05:02	30/04/2026	07:05 PM	1.15	0	253
30/04/2026, 19:16:50	30/04/2026	07:16 PM	4.47	0.549	983
30/04/2026, 19:16:55	30/04/2026	07:16 PM	4.68	0.759	1029
30/04/2026, 19:17:01	30/04/2026	07:17 PM	2.62	0	576
30/04/2026, 19:17:06	30/04/2026	07:17 PM	2.1	0	462
30/04/2026, 19:17:12	30/04/2026	07:17 PM	2	0	441
30/04/2026, 19:17:17	30/04/2026	07:17 PM	1.84	0	404
30/04/2026, 19:17:22	30/04/2026	07:17 PM	2.1	0	463
30/04/2026, 19:17:28	30/04/2026	07:17 PM	1.46	0	322
30/04/2026, 19:17:33	30/04/2026	07:17 PM	1.6	0	351

Telegram bot warning:



Challenges and Solutions

1-Accurate AC Measurement: Standard analog samples don't reflect true power consumption.
Solution: True RMS calculation (Squaring, Averaging, and Square Rooting)
across 2000 samples
2-Current Clamp Inaccuracy (Ambient Noise): External magnetic fields and environmental factors often caused the SCT-013 sensor to report non-zero readings even when the machine was off.
Solution: We implemented an ambient calibration function that baselines the sensor in its environment before operation, ensuring that only actual machine current is measured
3- Unauthorized Operation: Heavy machinery could be used without oversight or digital logging.
Solution: Developed an Event-Driven State Machine that triggers an alarm and a Telegram alert if current exceeds 1.0A without a successful NFC authentication

Conclusion

The Machine Guardian firmware provides a robust framework for enhancing industrial safety and accountability. By integrating physical security (NFC) with electrical monitoring (CT Sensors), it bridges the gap between manual operation and digital oversight.
The inclusion of an "Unauthorized" state and automated reporting features significantly reduces the risk of equipment misuse and helps management track precise energy costs per operator. Future iterations could expand on this by implementing MQTT for enterprise-level cloud integration or adding vibration sensors for predictive maintenance.